

📌 Part 1: Introduction to Logistic Regression

◆ What is Logistic Regression?

👉 Logistic Regression is a Machine Learning algorithm used for **Classification Problems**.

👉 It predicts:

- Yes / No
 - Pass / Fail
 - Spam / Not Spam
 - Disease / No Disease
-

📌 Part 2: Why Use Logistic Regression?

👉 When the output is a category instead of a number.

Examples:

Problem	Output
Pass Prediction	Pass/Fail
Spam Detection	Spam/Not Spam
Disease Detection	Positive/Negative

📌 Part 3: Theory Answers (IMPORTANT)

◆ 1. What is Logistic Regression?

👉 Logistic Regression is a supervised machine learning algorithm used for classification tasks.

◆ 2. Is Logistic Regression a Regression or Classification Algorithm?

👉 Despite the name "Regression", it is used for Classification.

◆ 3. What Type of Output Does Logistic Regression Produce?

👉 Category labels.

Examples:

- 0 = Fail
 - 1 = Pass
-

◆ 4. What is Binary Classification?

👉 Classification with only two classes.

Examples:

- Yes/No
 - True/False
 - Spam/Not Spam
-

◆ 5. What is a Class Label?

👉 A class label is the output category predicted by the model.

📌 Part 4: Example Dataset

Student Result Prediction

Study Hours Result

1	0
2	0
3	0
4	1
5	1

Study Hours Result

6	1
---	---

Meaning

Value Meaning

0	Fail
---	------

1	Pass
---	------

Part 5: Import Required Libraries

```
import pandas as pd
```

```
from sklearn.linear_model import LogisticRegression
```

Part 6: Create Dataset

```
data = {  
    "Hours": [1, 2, 3, 4, 5, 6],  
    "Result": [0, 0, 0, 1, 1, 1]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

Part 7: Features and Labels

```
X = df[["Hours"]]
```

```
y = df["Result"]
```

Explanation

 X = Input Feature

 y = Output Label

Part 8: Create Logistic Regression Model

```
model = LogisticRegression()
```

Part 9: Train the Model

```
model.fit(X, y)
```

What Happens Here?

 Model learns:

More study hours → Higher chance of pass

Part 10: Make Prediction

```
prediction = model.predict([[5]])
```

```
print(prediction)
```

Output

```
[1]
```

Meaning:

 Student will Pass

Part 11: Predict Multiple Students

```
prediction = model.predict([[2],[4],[7]])
```

```
print(prediction)
```

Possible Output

```
[0 1 1]
```

Meaning:

Hours Result

2	Fail
4	Pass
7	Pass

Part 12: Complete Program

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
```

```
data = {
    "Hours": [1, 2, 3, 4, 5, 6],
    "Result": [0, 0, 0, 1, 1, 1]
}
```

```
df = pd.DataFrame(data)
```

```
X = df[["Hours"]]
y = df["Result"]
```

```
model = LogisticRegression()
```

```
model.fit(X, y)
```

```
prediction = model.predict([[5]])
```

```
print("Prediction:", prediction)
```

Part 13: Understanding Probability

Logistic Regression works using probabilities.

Example:

Probability Result

0.20	Fail
------	------

Probability Result

0.40	Fail
0.80	Pass
0.95	Pass

Get Probability

```
model.predict_proba([[5]])
```

Example Output:

```
[[0.15 0.85]]
```

Meaning:

- Fail Probability = 15%
- Pass Probability = 85%

Part 14: Real-World Applications

Education

 Pass/Fail Prediction

Healthcare

 Disease Detection

Banking

 Loan Approval

Email Services

 Spam Detection

E-Commerce

👉 Customer Purchase Prediction

📌 Part 15: Visualization

Study Hours

1 — Fail

2 — Fail

3 — Fail

4 — Pass

5 — Pass

6 — Pass

📌 Part 16: Logistic Regression Workflow

Dataset



Feature (X)



Label (y)



Train Model



Prediction



Pass / Fail

📌 Part 17: Difference Between Linear and Logistic Regression

Linear Regression

Logistic Regression

Predicts Numbers

Predicts Categories

Linear Regression

Continuous Output

Salary Prediction

House Price Prediction

Logistic Regression

Discrete Output

Pass/Fail Prediction

Tasks for Day 38

Task 1: Theory

1. What is Logistic Regression?
 2. Why is it used?
 3. What is Binary Classification?
 4. What is a Class Label?
 5. Difference between Linear and Logistic Regression?
-

Task 2: Dataset Creation

1. Create Pass/Fail dataset
 2. Create Disease dataset
 3. Create Spam dataset
-

Task 3: Model Training

1. Train Logistic Regression model
 2. Use fit()
 3. Predict outputs
-

Task 4: Prediction Practice

1. Predict student result
2. Predict disease status
3. Predict spam message status

 Task 5: Probability Analysis

1. Use `predict_proba()`
2. Explain output
3. Compare probabilities

 Task 6: Classification Logic

For each problem identify whether Logistic Regression can be used:

1. Pass/Fail Prediction
2. House Price Prediction
3. Spam Detection
4. Disease Prediction
5. Salary Prediction